Properties of the z-Transform

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Key Concepts

1) The properties of the z-transform are analogous to those of the discrete-time Fourier transform.

2) The z-transform of a weighted sum of signals is the weighted sum of the individual signal z-transforms. The region of convergence (ROC) is the intersection of the individual ROCs.

3) The z-transform of the convolution of two signals is the product of the individual signal z-transforms. The ROC is the intersection of the individual ROCs.

4) Multiplication of $x[n]$ by n has z-transform given by $-z$ times the derivative of $X(z)$ with respect to z. The ROC is unchanged.

5) Multiplication of $x[n]$ by the exponential sequence z_0^n has z-transform given by $X\left(\frac{z}{z_0}\right)$. The ROC the original ROC multiplied by $|z_0|$.

6) Time-shifting $x[n]$ by n_0 samples multiplies $X(z)$ by z^{-n_0}. The ROC is unchanged except for the addition or deletion of $z = 0$ or $z = \infty$.

7) The z-transform of the impulse response of a system is called the system function.

8) The system function for a system described by a linear constant coefficient difference equation is a ratio of polynomials in z^{-1}. The coefficients of the polynomials are given by the coefficients of the difference equation.

Video Screenshots

z-transform properties are analogous to DTFT properties

1) Linearity

$$ax_1[n] + bx_2[n] \overset{z}{\longleftrightarrow} aX_1(z) + bX_2(z) \qquad R_{x_1} \cap R_{x_2}$$

2) Convolution

$$x_1[n] * x_2[n] \overset{z}{\longleftrightarrow} X_1(z)X_2(z) \qquad R_{x_1} \cap R_{x_2}$$



$$Y(z) = X(z)H(z) \qquad R_x \cap R_H$$

system function
transfer function

3) Differentiation of X(z)

$$n\,x[n] \overset{z}{\longleftrightarrow} -z\frac{d}{dz}X(z) \qquad \text{ROC } R_x$$

$$n a^n u[n] \overset{z}{\longleftrightarrow} -z\frac{d}{dz}\left(\frac{1}{1-az^{-1}}\right) = \frac{az^{-1}}{(1-az^{-1})^2}, \quad |z|>|a|$$

4) Multiplication by exponential sequence

$$z_0^n\,x[n] \overset{z}{\longleftrightarrow} X(z/z_0) \qquad \text{ROC } |z_0|\cdot R_x$$

if $X(z)$ has a pole/zero at α —

$X(z/z_0)$ has a pole/zero at $z_0\alpha$

$z_0 = re^{j\phi} \Rightarrow z_0\alpha = r|\alpha|e^{j\arg\{\alpha\}+\phi}$.

scale radius by r, shift angle by ϕ

4:40

5) Time-shift property

$$x[n-n_0] \xleftrightarrow{z} z^{-n_0} X(z) \quad \text{ROC } R_x \text{ (except possibly } z=0 \text{ or } z=\infty)$$

System described by difference equation

$$y[n] + a_1 y[n-1] + \cdots a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \cdots + b_M x[n-M]$$

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

$$\sum_{k=0}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$Y(z) = \underbrace{\frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}}_{H(z)} X(z)$$

$H(z)$ system/transfer function



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